MMML `cli` tutorial

This walkthrough follows the numbered scripts in `mmml_tutorial/cli/`: build a small system with PyCHARMM, run GPU DFT and sampling, label data, split into training NPZs, train PhysNet (optional), and train the joint PhysNet+DCMNet model with ESP grids. Default residue is `BENZ` (`shared.source`); override with `RES=TIP3` (or another residue) before sourcing if you want a different chemistry.

Each step below links a **figure** (molecule, plot, or schematic) and a **log excerpt** from the checked-in `*.log` files in `cli/`, so a prepared tutorial room can show expected console output without re-running long jobs.

For a flat command index, see `cli_cheatsheet.typ`.

MMML

https://github.com/EricBoittier/mmml

mmml_tutorial

https://github.com/EricBoittier/mmml_tutorial

Setup: `uv`, MMML, PyCHARMM, and Packmol

Install `uv` once per machine. The standalone installer is simplest; `pip` or `conda` are fine when those tools are already managed on the machine:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
export PATH="$HOME/.local/bin:$PATH"
uv --version

# Alternatives
python -m pip install --user uv
conda install -c conda-forge uv
```

Clone and install MMML into a project-managed environment. The package currently targets Python 3.13. Prefer the full extra for tutorial machines:

```
git clone https://github.com/EricBoittier/mmml.git
cd mmml
uv python install 3.13
uv sync --extra all
uv run mmml --help
```

Choose a narrower full extra only when the hardware requires it:

```
# CPU-only machines or laptops
uv sync --extra all-cpu

# CUDA 12 nodes
uv sync --extra all-cuda12
```

For editable development from an already active environment, the equivalent install is:

```
uv pip install -e ".[all]"
```

Compile CHARMM for PyCHARMM

MMML's system-building and hybrid-MD paths import PyCHARMM and expect CHARMM to be built as a shared library. From a licensed CHARMM source tree:

```
cd /path/to/charmm
./configure --as-library
make -j "$(nproc)"
```

Then point MMML at that build with a `CHARMMSETUP` file in the MMML repository root:

```
cat > CHARMMSETUP <<'EOF'
CHARMM_HOME=/path/to/charmm
CHARMM_LIB_DIR=/path/to/charmm/build/cmake
EOF
```

Quick check:

```
uv run python - <<'PY'
from mmml.interfaces.pycharmmInterface.import_pycharmm import CHARMM_HOME, CHARMM_LIB_DIR
print("CHARMM_HOME", CHARMM_HOME)
print("CHARMM_LIB_DIR", CHARMM_LIB_DIR)
PY
```

If a cluster build segfaults during CHARMM energy display calls, run tutorial commands with `SKIP_CHARMM_ENERGY_SHOW=1` or pass `--skip-energy-show` where the CLI exposes it.

Install or compile Packmol

`mmml make-box` shells out to the `packmol` executable. Install the binary with conda when possible:

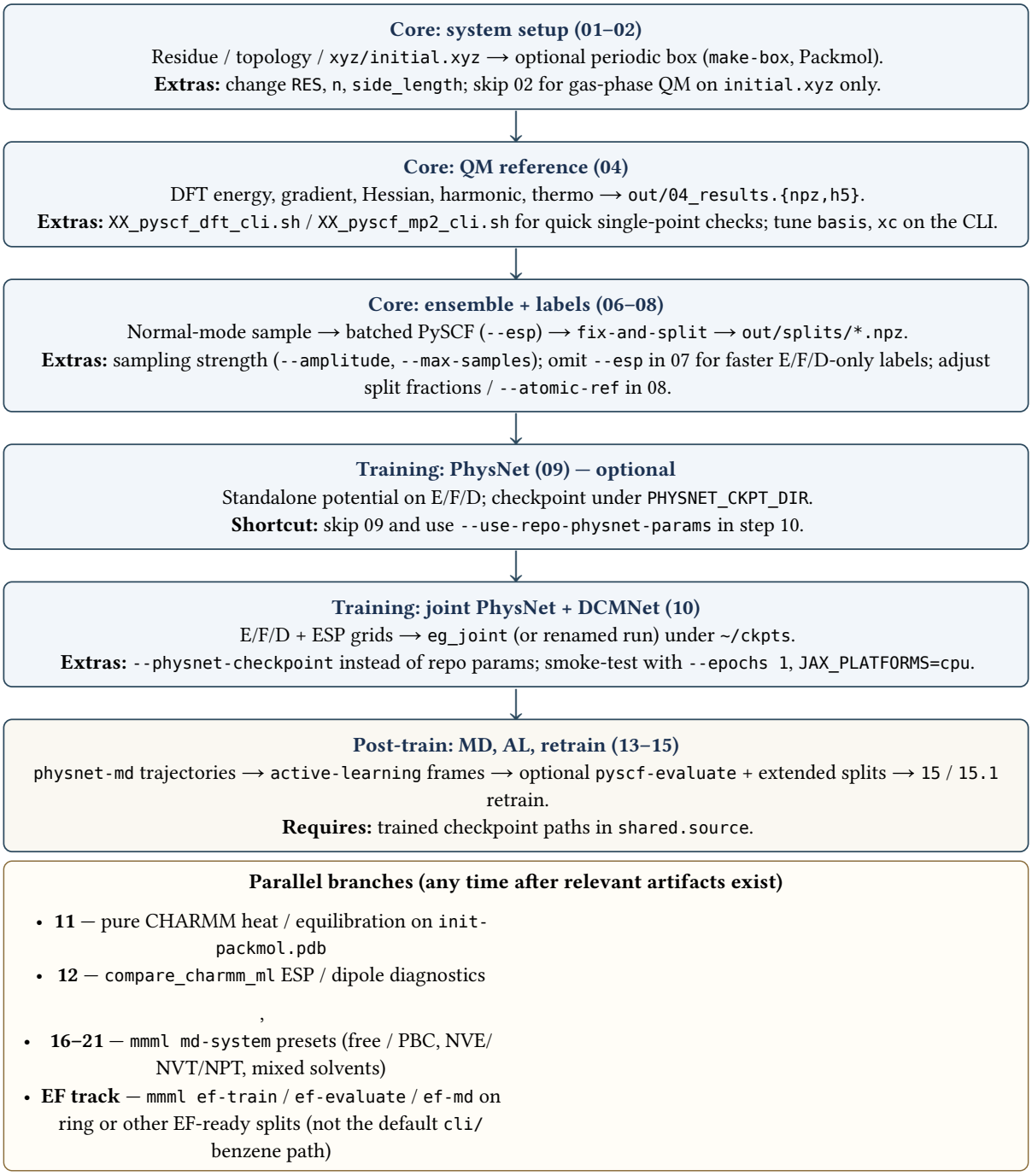
```
conda install -c conda-forge packmol
command -v packmol
```

If conda is not available, compile Packmol from source and put the executable on `PATH`:

```
git clone https://github.com/m3g/packmol.git
cd packmol
make
export PATH="$PWD:$PATH"
command -v packmol
```

Tutorial map: core path, options, and extras

The figure below is the **contract** for the rest of the document: a linear **core** pipeline (stacked boxes) you can trim for demos, plus **optional branches** (bottom panel) for classical MM, diagnostics, neural MD, active learning, preset `md-system` workloads, and equivariant field (EF) models.



Listing 1: End-to-end **tutorial coverage** (center column) versus **extra functionality** you can demonstrate with the same repo (branches). Numbers refer to scripts in `mmml_tutorial/cli/`.

How to choose options at each stage

- **Residue and box (01–02).** Use `shared.source` or `RES=…` for chemistry. Need a condensed phase or interface? Increase `--n` and box `side_length`, or add solvent flags if your `make-box` invocation supports them. Gas-phase dataset: run 01 only and point step 04 at `xyz/initial.xyz`.
- **Reference QM (04).** The tutorial script requests Hessian + harmonic for normal-mode sampling. For a **cheap** sanity check, use the `XX_*` scripts or a minimal `pyscf-dft` run without `--hessian`. Production datasets usually keep harmonic data for 06.
- **Sampling (06).** `--max-samples` trades cost (step 07) against diversity. Multiple `--amplitudes` are supported by the CLI if you customize the script.

- **Batch evaluation (07).** `--esp` adds cost and ESP grid tensors required for DCMNet. For pure PhysNet-on-EF training without ESP, you can drop `--esp` and adjust step 08 / 10 inputs accordingly.
- **Splits (08).** `fix-and-split` chooses train/valid/test fractions and optional atomic energy references. For **active learning**, merge new evaluated NPZs into extended directories (`splits_extended*`) as in steps 14–15.
- **PhysNet vs joint (09–10).** If the portable repo PhysNet matches your stoichiometry, prefer `--use-repo-physnet-params` and skip long 09 runs in the classroom. Use `--physnet-checkpoint` when you need your own step-09 run.
- **After training.** Step 13 drives exploratory MD; step 14 subsamples frames for re-QM. Steps 16–21 showcase `md-system` automation (cutoff figures in the tutorial assets). The EF subsection documents a **different** model family on its own NPZ layout.

Figures and log excerpts

- Regenerate PNGs under `assets/cli/` with `uv run python generate_tutorial_assets.py` from `mmml_tutorial/cli/`.
- Log snippets are illustrative; your exact timings and energies will differ.

Recent CLI updates

- Joint training: `--use-repo-physnet-params` bootstraps PhysNet from portable repo JSON (no fresh PhysNet run required for a DCMNet demo).
- EF training: `--rot-augment`, `--rot-perturbation`.
- Checkpoints: `mmml extract-checkpoint-metrics <run_dir> -o metrics.png --log-loss`.

What this tutorial covers

See **Tutorial map** (above) for the full pipeline and branches. In short: scripts `01–08` build the ML dataset; `09–10` train; `11–21` and EF commands are optional extensions using the same CLI.

Live tutorial strategy

1. Walk through steps 01–02 and 04 conceptually (show figures + logs).
2. Run or show 06–07 only if GPUs are ready; otherwise use precomputed `out/`.
3. For DCMNet, either run `08` then `10`, or supply prepared `out/splits/*.npz`.
4. Smoke-test joint training: `--epochs 1`, `--ckpt-dir /tmp/mmml_ckpts`, `JAX_PLATFORMS=cpu` on laptops.

Quickest joint-training demo (after splits exist):

```
cd mmml_tutorial/cli
bash 10_physnet_dcmnet_train_cli.sh
```

Environment checklist

- Work from `mmml_tutorial/cli` (paths below are relative to that directory).
- Install `uv`, run `uv python install 3.13`, and prefer `uv sync --extra all` from the MMML repo.
- PyCHARMM + Packmol for steps 01–02; PySCF/GPU stack for 04 and 07.
- Compile CHARMM with `./configure --as-library`, then set `CHARMM_HOME` and `CHARMM_LIB_DIR` in `CHARMMSETUP`.
- Install Packmol with `conda install -c conda-forge packmol` or compile it from source and add it to `PATH`.
- `JAX_PLATFORMS=cpu` if CUDA init breaks before argparse.

Directory map (`mmml_tutorial/cli/`)

- `shared.source` — `RES`, PhysNet trainer path, checkpoint dirs, NPZ paths, MD defaults.
- `01_make_res_cli.sh` — residue/topology + `xyz/initial.xyz`.
- `02_make_box_cli.sh` — Packmol box `pdb/init-packmol.pdb`.
- `04_pyscf_dft_cli_full.sh` — DFT + Hessian + harmonic + thermo ⟶ `out/04_results.{npz,h5}`.
- `06_normal_mode_sample_cli.sh` — `out/06_sampled.npz`.
- `07_pyscf_evaluate_cli.sh` — batched QM + ESP ⟶ `out/07_evaluated*.npz`.
- `08_fix_and_split_cli.sh` — splits under `out/splits/`.

- `09_physnet_train_cli.sh` — PhysNet (trainer script from `shared.source`).
- `10_physnet_dcmnet_train_cli.sh` — joint PhysNet+DCMNet → `~/ckpts/eg_joint/…`.
- `11_run_pycharmm_cli.sh` — classical heat/equilibration.
- `12_charmm_esp_dipole_comparison.sh` — CHARMM vs ML diagnostics.
- `13_physnet_md.sh` — PhysNet MD trajectories.
- `14_active_learning.sh` — frames for re-labeling.
- `15_physnet_retrain_cli.sh`, `15.1_physnet_retrain_cli.sh` — continued training.
- `16_md_10mer_free_nve.sh` … `20_md_10mer_pbc_npt.sh` — `mmml md-system` presets.
- `21_md_system_meoh_tip3_1to1.sh` — mixed composition example.
- `run_full_training.sh` — 06⟶10 automation with shortened epochs in-repo.

Data flow

```
xyz/initial.xyz
  -> out/04_results.{npz,h5}
  -> out/06_sampled.npz
  -> out/07_evaluated*.npz
  -> out/splits/{energies_forces_dipoles,grids_esp}_{train,valid,test}.npz
  -> ~/ckpts/<PHYSNET_NAME>/   (step 09, optional)
  -> ~/ckpts/eg_joint/         (step 10, Orbax run id may include a UUID suffix)
```

Recommended run order

1. `01_make_res_cli.sh`
2. `02_make_box_cli.sh`
3. `04_pyscf_dft_cli_full.sh`
4. `06_normal_mode_sample_cli.sh`
5. `07_pyscf_evaluate_cli.sh`
6. `08_fix_and_split_cli.sh`
7. `09_physnet_train_cli.sh` **(optional if using repo PhysNet params in step 10)**
8. `10_physnet_dcmnet_train_cli.sh`
9. `13_physnet_md.sh`, `14_active_learning.sh`, `15*.sh` as needed
10. `16`–`21` MD presets for `md-system` exploration

Side branches: `11_run_pycharmm_cli.sh`, `12_charmm_esp_dipole_comparison.sh`.

Step-by-step notes

Step 01: Make residue

Purpose: CHARMM residue/topology and `xyz/initial.xyz` for the chosen `RES`.

```
cd mmml_tutorial/cli
bash 01_make_res_cli.sh
```

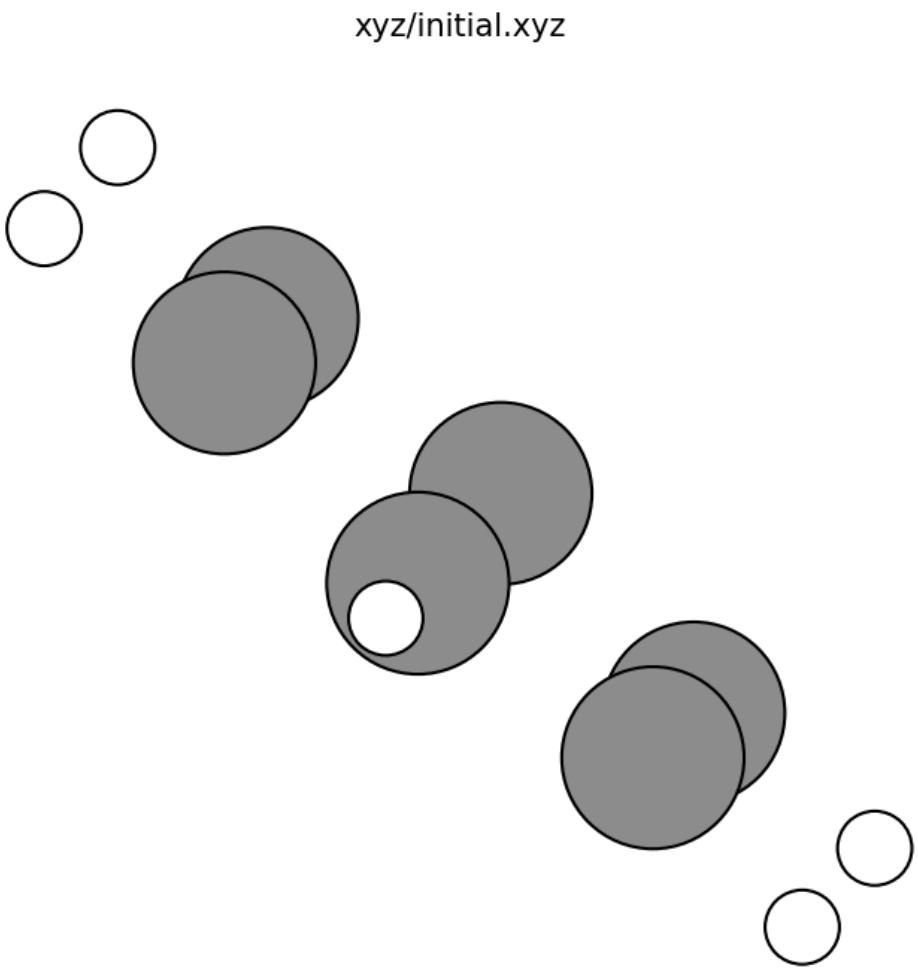


Figure 1: Study system (`BENZ` by default): ASE `plot_atoms` on `xyz/initial.xyz` (asset script falls back to built-in C6H6 if missing).

Main outputs: `pdb/initial.pdb`, `psf/initial.psf`, `xyz/initial.xyz`, `out/01_last_command.txt`.

```
=== 01: make_res (CLI) ===
Command: mmml make-res --res BENZ --skip-energy-show
...
                    NORMAL TERMINATION BY NORMAL STOP
Saved command: out/01_last_command.txt
```

Listing 2: Tail of `mmml_tutorial/cli/01.log` (CHARMM output abbreviated).

Step 02: Build box

Purpose: Pack copies into a cubic cell for classical equilibration or visualization.

```
bash 02_make_box_cli.sh
```

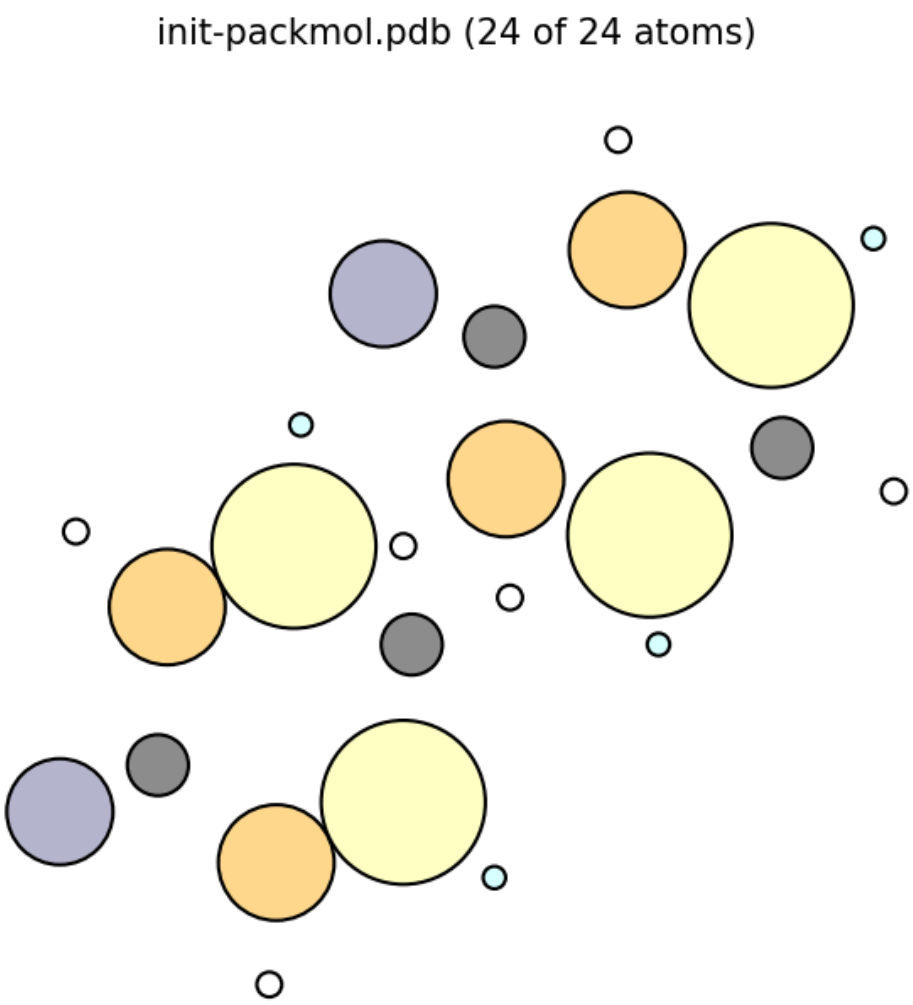


Figure 2: ASE `plot_atoms` on a subset of `pdb/init-packmol.pdb` (large cells are truncated for clarity).

```
=== 02: make_box (CLI) ===
Command: mmml make-box --res BENZ --n 2 --side_length 25.0
...
Saved command: out/02_last_command.txt
```

Listing 3: Excerpt from `cli/02.log`.

Step 04: Full DFT with PySCF

Purpose: reference energy, gradient, Hessian, harmonic analysis, and thermo on `xyz/initial.xyz`.

```
bash 04_pyscf_dft_cli_full.sh
```

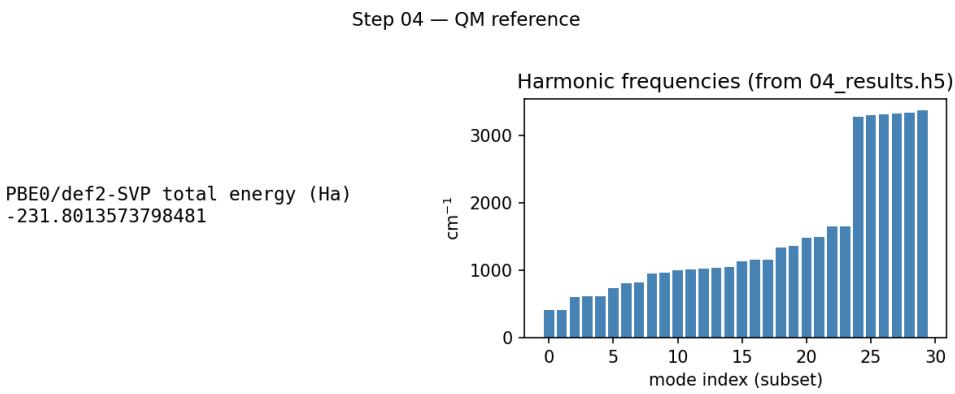


Figure 3: DFT total energy (from `04.log`) and harmonic frequencies when `out/04_results.h5` is present.

Main outputs: `out/04_results.npz`, `out/04_results.h5`, `pyscf.log`.

```
=== 04: DFT full (energy, gradient, hessian, harmonic, thermo) ===
Command: uv run mmml pyscf-dft --mol xyz/initial.xyz --energy --gradient --hessian --harmonic --thermo
--output out/04_results
...
total energy = -231.8013573798481
...
Results saved to out/04_results.npz (ML keys) and out/04_results.h5 (all arrays)
Elapsed: 16.68 s
```

Listing 4: Excerpt from `cli/04.log`.

Step 06: Normal-mode sampling

Purpose: build a local geometry ensemble from the harmonic model.

```
bash 06_normal_mode_sample_cli.sh
```

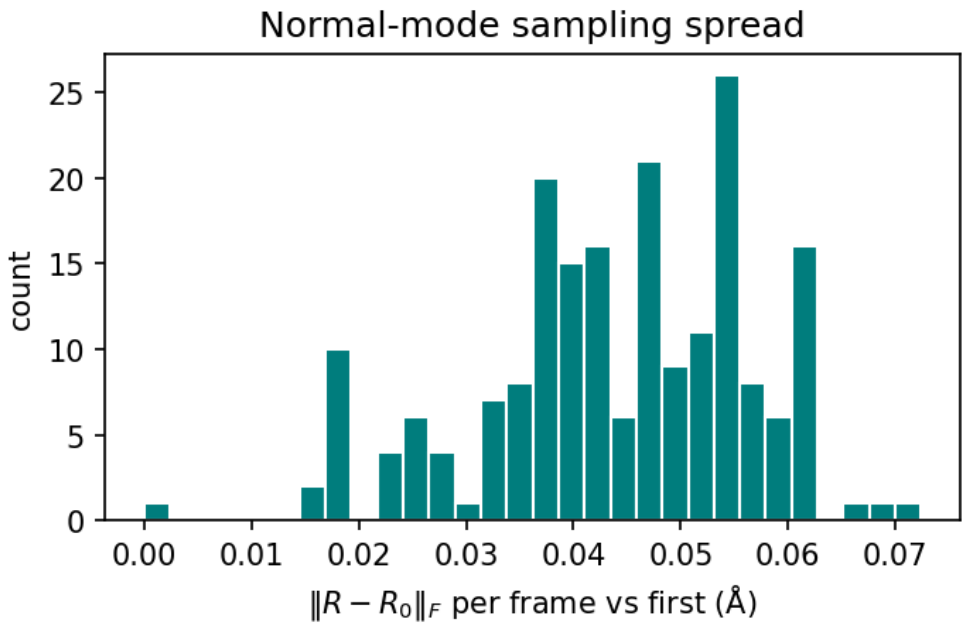


Figure 4: Distribution of displacements relative to the first sampled frame (`out/06_sampled.npz`).

```
=== 06: Normal mode sampling ===
Command: mmml normal-mode-sample -i out/04_results.h5 -o out/06_sampled.npz --amplitude 0.1 --max-
samples 200
Saved 200 geometries to out/06_sampled.npz
Elapsed: 0.03 s
```

Listing 5: Excerpt from `cli/06.log`.

Step 07: Evaluate sampled geometries

Purpose: batch QM labels and ESP grids for ML.

```
bash 07_pyscf_evaluate_cli.sh
```

If `out/07_evaluated.npz` already exists, the CLI may write `out/07_evaluated_2.npz` (see log). Step 08 picks the newest `out/07_evaluated*.npz`.

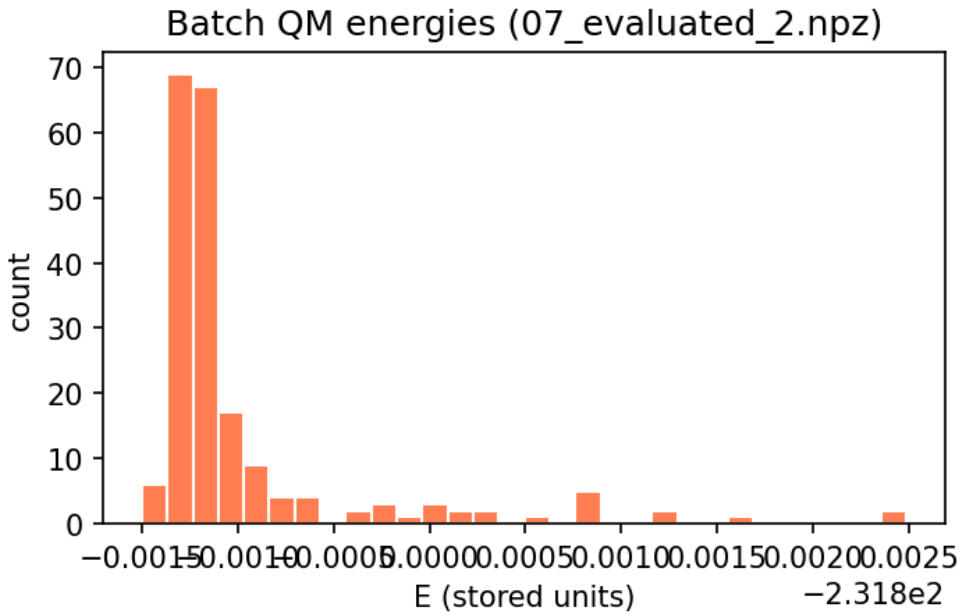


Figure 5: Histogram of total energies across the evaluated batch (`out/07_evaluated*.npz`).

```
=== 07: pyscf-evaluate (energy, forces, dipoles, ESP) ===
Command: uv run mmml pyscf-evaluate -i out/06_sampled.npz -o out/07_evaluated.npz --esp
Evaluating 200 geometries with pyscf-dft (GPU)...
Output exists, using out/07_evaluated_2.npz instead of out/07_evaluated.npz
Saved to out/07_evaluated_2.npz
  R: (200, 12, 3), E: (200,), F: (200, 12, 3)
Elapsed: 502.56 s
```

Listing 6: Excerpt from `cli/07.log`.

Step 08: Fix units and split datasets

Purpose: atomic references, unit cleanup, deterministic train/valid/test NPZs.

```
bash 08_fix_and_split_cli.sh
```

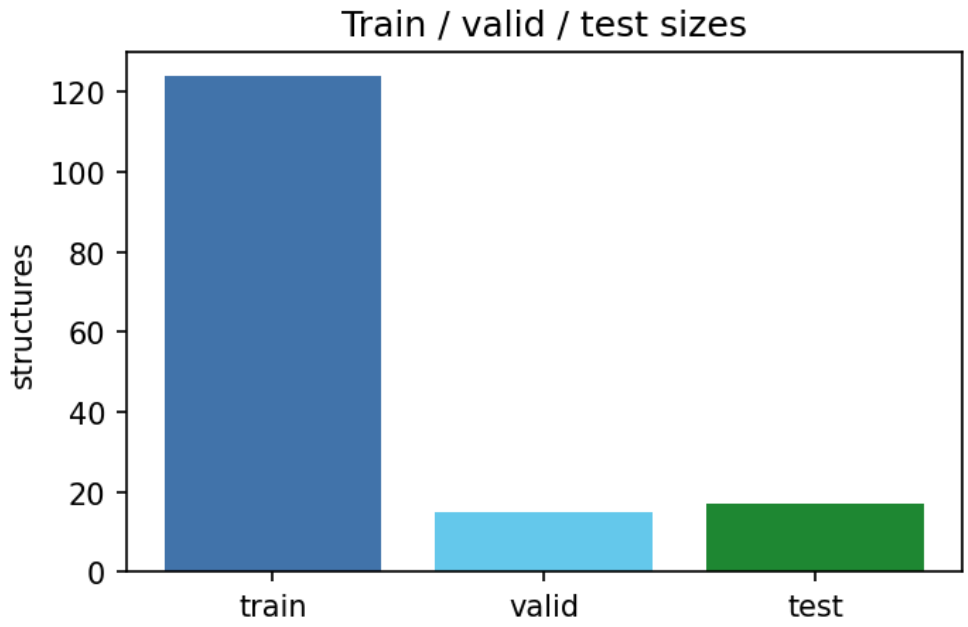


Figure 6: Structure counts in `out/splits/` after `fix-and-split`.

```
=== 08: fix-and-split (train/valid/test) ===
Command: mmml fix-and-split --efd out/07_evaluated_2.npz --output-dir out/splits
```

Listing 7: The shell script passes the newest matching `out/07_evaluated` NPZ (see `08_fix_and_split_cli.sh`).

Typical outputs:

- `out/splits/energies_forces_dipoles_{train,valid,test}.npz`
- `out/splits/grids_esp_{train,valid,test}.npz`

Step 09: Train PhysNet

Purpose: baseline potential on E/F/D (charges + ZBL in the bundled trainer invocation).

```
bash 09_physnet_train_cli.sh
```

Checkpoints land under `"$PHYSNET_CKPT_DIR/$PHYSNET_NAME/"` (see `shared.source`).

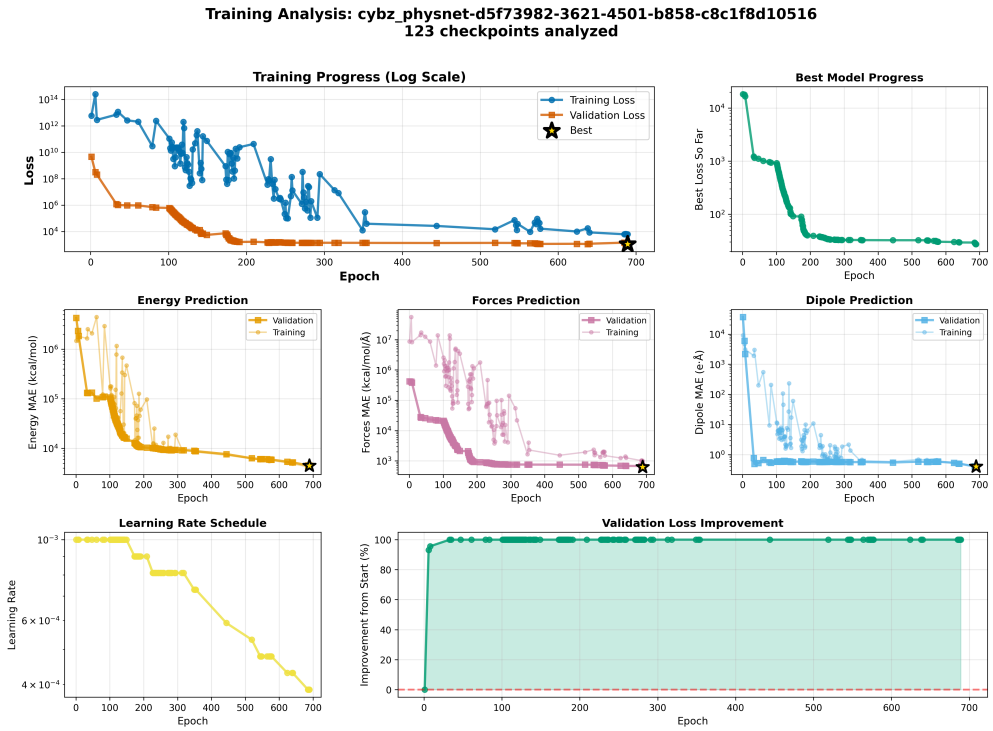


Figure 7: Orbax metrics from `mmml extract-checkpoint-metrics` when a PhysNet run directory is available; otherwise regenerate assets with placeholders.

```
mmml extract-checkpoint-metrics "$PHYSNET_CKPT_DIR/$PHYSNET_NAME" \
  -o typst_docs/tutorial/assets/cli/step09_training_metrics.png --log-loss
```

(Adjust the output path if you run the command from another working directory.)

Step 10: Joint PhysNet + DCMNet training

Purpose: co-train on energies, forces, dipoles, and ESP grids.

```
bash 10_physnet_dcmnet_train_cli.sh
```

Equivalent command (from the script):

```
python -m mmml.cli.misc.train_joint \
  --train-efd out/splits/energies_forces_dipoles_train.npz \
  --train-esp out/splits/grids_esp_train.npz \
  --valid-efd out/splits/energies_forces_dipoles_valid.npz \
  --valid-esp out/splits/grids_esp_valid.npz \
  --use-repo-physnet-params \
  --epochs 1000 \
  --batch-size 1 \
  --name eg_joint \
  --ckpt-dir ~/ckpts --plot-results --plot-freq 0
```

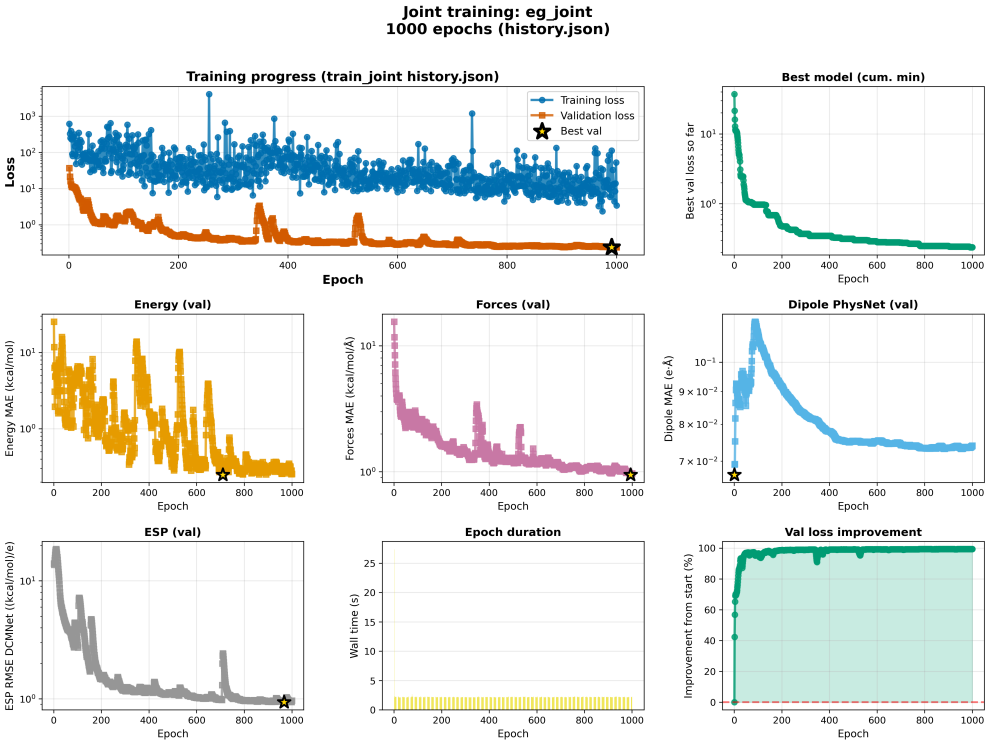


Figure 8: Joint training metrics; generate with `extract-checkpoint-metrics` on `~/ckpts/eg_joint*`.

Notes:

- Use only one of `--use-repo-physnet-params`, `--physnet-checkpoint`, or `--restart`.
- Smoke test: `--epochs 1`, `--ckpt-dir /tmp/mmml_ckpts`.

EF training: rotational augmentation example

```
mmml ef-train \
  --train-npz out/splits_ef0_ring/energies_forces_dipoles_train.npz \
  --valid-npz out/splits_ef0_ring/energies_forces_dipoles_valid.npz \
  --test-npz out/splits_ef0_ring/energies_forces_dipoles_test.npz \
  --rot-augment --rot-perturbation 1.0 \
  --clip_norm 1000 \
  --output-dir ~/ckpts/ring_ef_ptT_run2 \
  --num_iterations 2 --max_degree 2
```

(EF ring dataset paths are examples; adapt to your splits.)

Step 11: Pure CHARMM equilibration

```
bash 11_run_pycharmm_cli.sh
```

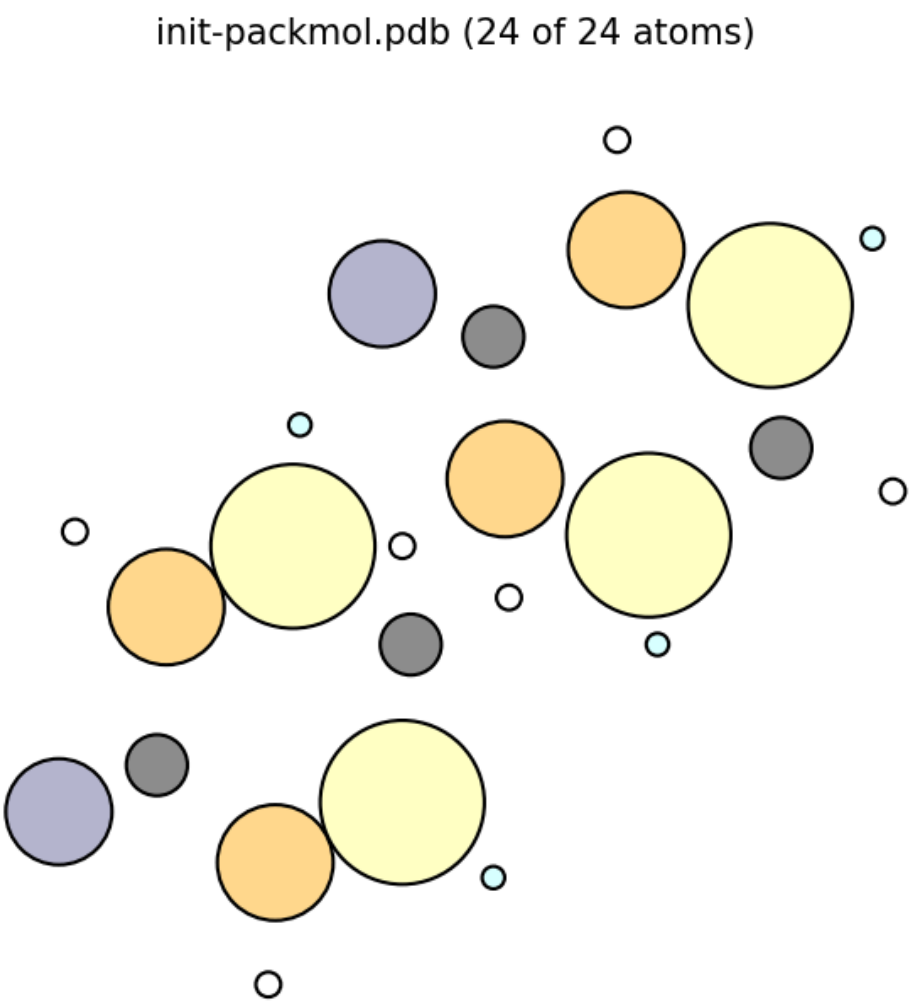


Figure 9: Same packed cell as step 02: this is the geometry family fed to `run-pycharmm` before heat/equilibration outputs are written.

Outputs: `pdb/{heat,equi}.pdb`, `dcd/{heat,equi}.dcd`, `res/{heat,equi}.res`.

Step 12: CHARMM vs ML comparison

```
bash 12_charmm_esp_dipole_comparison.sh
```

The script runs `compare_charmm_ml` against `~/ckpts/eg_joint` and `out/splits/*_test.npz`; edit the checkpoint path if your joint run uses another directory.

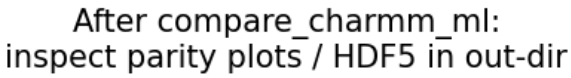


Figure 10: Placeholder: replace by parity or GUI screenshots after you run `compare_charmm_ml` and regenerate assets if desired.

Step 13: PhysNet MD sampling

```
bash 13_physnet_md.sh
```

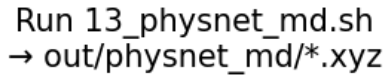


Figure 11: ASE `plot_atoms` on the last frame of ASE or JAX-MD `xyz` under `out/physnet_md/` when trajectories exist.

Step 14: Active learning extraction

```
bash 14_active_learning.sh
```

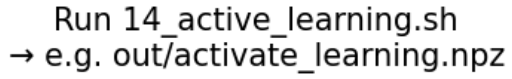


Figure 12: Spread of geometries in the active-learning NPZ (`out/activate_learning.npz` by default in `shared.source`).

Suggested follow-up:

```
mmml pyscf-evaluate -i "$ACTIVE_LEARNING_OUT" -o out/md_evaluated.npz --esp
mmml fix-and-split --efd "$PHYSNET_TRAIN_NPZ" out/md_evaluated.npz -o out/splits_extended
```

Step 15 / 15.1: Retraining on extended splits

Continue PhysNet training with `--restart` pointing at your previous run; see `15_physnet_retrain_cli.sh` and `15.1_physnet_retrain_cli.sh` for one or multiple extension directories (`splits_extended/`).

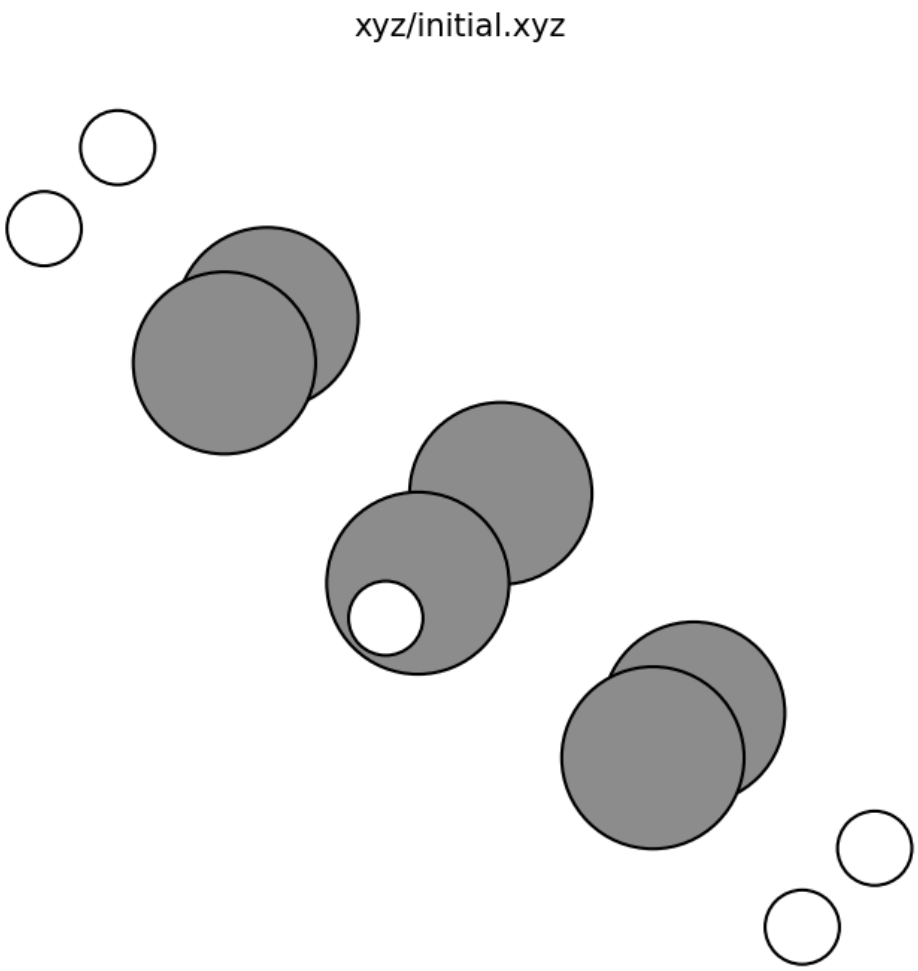


Figure 13: Chemistry reminder: extended datasets still target the same `RES` / atom count configured in `shared.source`.

`mmml md-system` cutoff schematics (steps 16–19)

These scripts call `mmml md-system` with different ensembles. Logs record a complementary cutoff schematic PNG; copies are placed in `assets/cli/` when the generator finds the source files.

```
bash 16_md_10mer_free_nve.sh
bash 17_md_10mer_free_nvt.sh
bash 18_md_10mer_pbc_nve.sh
bash 19_md_10mer_pbc_nvt.sh
```

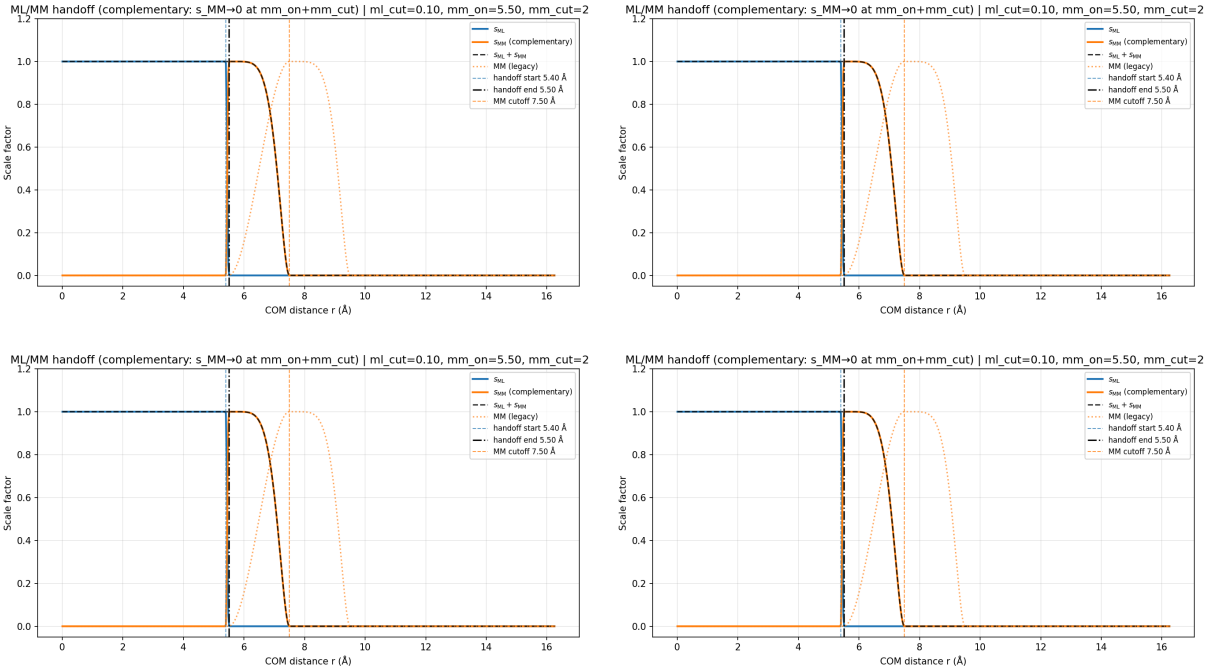


Figure 14: Cutoff layout figures referenced from `16.log`–`19.log` (paths may differ on your machine).

Periodic NPT and mixed-solvent examples: `20_md_10mer_pbc_npt.sh`, `21_md_system_meoh_tip3_1to1.sh`. The `md-system` wrapper has an explicit `--backend auto|ase|jaxmd` selector: `auto` keeps ASE for the free/PBC NVE/NVT presets and routes `pbc_npt` through JAX-MD. Use `--backend jaxmd` when you want the periodic NVE/NVT presets to use the JAX-MD runner too.

Generated molecules are placed at random 3D COM positions rather than on a single plane. Use `--seed` (or `MDSYS_SEED` in the tutorial scripts) to reproduce or reshuffle placement; `--spacing` controls the target minimum COM spacing. Periodic presets choose a cubic box automatically from the initial cluster extent plus padding. Override it with `--box-size <angstrom>` or, in the tutorial shell scripts, with `MDSYS_BOX_A`:

```
MDSYS_PS=10 MDSYS_N_MOLECULES=100 MDSYS_BOX_A=60 MDSYS_SEED=7 bash 20_md_10mer_pbc_npt.sh
```

JAX-MD preset runs now do a staged pre-MD minimization. CHARMM SD/ABNR runs first, including an energy/force warmup with the internal force-field terms active. The MMML calculator then runs ASE BFGS to `fmax=0.1`; if BFGS does not reach that threshold, ASE FIRE is used before the JAX-MD minimizer starts. The optimizer target remains `0.1`, while the default abort threshold is looser at `--max-fmax-after-min 2.0`. The best force and best energy structures from this ASE pre-minimization are saved, and the JAX-MD handoff continues from the best-force structure. Tune these lower-level knobs by putting `--extra-args` last, for example:

```
mmml md-system --setup pbc_npt --backend jaxmd --extra-args --pre-min-steps 200 --fire-min-steps 500
```

JAX-MD periodic runs record every 100 steps by default (`--steps-per-recording 100`). Fixed-box NVT/NVE neighbor reuse defaults to `--jax-md-update-interval 5` with a `0.2 A` skin; NPT keeps the safer every-recording-block refresh.

Optional scripts (`XX_*`)

- `XX_pyscf_dft_cli.sh` — minimal DFT energy demo.
- `XX_pyscf_mp2_cli.sh` — MP2 reference example.

`shared.source` variables you will likely edit

- `RES`, `PHYSNET_TRAINER`, `PHYSNET_CKPT_DIR`, `PHYSNET_NAME`, `PHYSNET_CHECKPOINT`
- `PHYSNET_NATOMS`, `PHYSNET_EPOCHS`, `PHYSNET_BATCH`
- `PHYSNET_TRAIN_NPZ`, `PHYSNET_VALID_NPZ`
- `PHYSNET_MD_OUT`, `PHYSNET_MD_TRAJ`, `ACTIVE_LEARNING_OUT`

- `MDSYS_*` knobs for `md-system` demos

Fast start options

Option A: bundled full script

```
cd mmml_tutorial/cli
bash run_full_training.sh
```

(Still runs expensive QM; shorten epochs/checkpoints inside the script for demos.)

Option B: manual progression

```
cd mmml_tutorial/cli
bash 01_make_res_cli.sh
bash 02_make_box_cli.sh
bash 04_pyscf_dft_cli_full.sh
bash 06_normal_mode_sample_cli.sh
bash 07_pyscf_evaluate_cli.sh
bash 08_fix_and_split_cli.sh
bash 10_physnet_dcmnet_train_cli.sh
```

Option C: joint smoke test (splits prepared)

```
cd mmml_tutorial/cli
JAX_PLATFORMS=cpu uv run python -m mmml.cli.misc.train_joint \
  --train-efd out/splits/energies_forces_dipoles_train.npz \
  --train-esp out/splits/grids_esp_train.npz \
  --valid-efd out/splits/energies_forces_dipoles_valid.npz \
  --valid-esp out/splits/grids_esp_valid.npz \
  --use-repo-physnet-params \
  --epochs 1 \
  --batch-size 1 \
  --name smoke_joint \
  --ckpt-dir /tmp/mmml_ckpts \
  --plot-freq 0
```

Caveats

- Script `echo` lines are sometimes shortened; the shell body is authoritative.
- `pyscf-evaluate` may append `_2`, `_3`, … when outputs exist; step 08 always consumes the newest file.
- GPU JAX failures before argparse: prefix with `JAX_PLATFORMS=cpu`.
- Portable PhysNet JSON must match the architecture expected by `train_joint` for your system size.

Troubleshooting

Joint training parameter merge errors

Healthy startup usually prints:

```
Loaded PhysNet config from checkpoint
Merging PhysNet params into joint model
Pre-computing edge lists
```

PySCF or CHARMM too slow for live sessions

Precompute `out/04_results.h5`, `out/06_sampled.npz`, `out/07_evaluated*.npz`, and `out/splits/*.npz`, then teach from logs + figures and run only short training.

Appendix: water-dimer or other dedicated layouts

Some course materials use a water-dimer directory with `out/splits_dimer/` and different `train_joint` names. The workflow is the same at the CLI level; only paths and `NATOMS` change. Start from the appropriate `shared.source` overrides for that dataset.

Next additions

- Wall-time table on reference hardware.
- Screenshots from `mmml gui` on trajectories produced in step 13.